

Hardware IP Card

Basic Information		
Licensing		Closed-source proprietary
Target FPGA or ASIC		XC7K325T, ASIC
Design Language		SystemVerilog
System-Level Features		
Coupling Type		Tightly-coupled
Data Types		
ISA		
Peak Ops Per Cycle		
Interfaces		
Supported Operations		
Compiler Support		Yes, specialized compiler based on LLVM17
External Memory Required		
Cache Or Scratchpad		
Architecture		
Type		CPU extension
Num Clock Domains		1
Control Interface	ISAExtension	Yes (RV32I; complies partly with Zicfiss extension)
Control Interface	Memory Mapped	No
Control Interface	Protocol	
Memory Interface	Interface Protocol	
Memory Interface	Hierarchy Level	
Memory Interface	Cached	
Memory Interface	IOMMU	
Interrupts		
Microarchitecture		
Parametrization	Parametric Num Units	No
Parametrization	Parametric Config	Yes
Programmability	Programmable Cores	No
Programmability	ISAExtension	
Software		
Compiler	Requires Specialized Compiler	Yes
Compiler	Supported Compilers	Specialized compiler based on LLVM17
Hardware Abstraction Layer	HAL Functions	No
highLevelAPI	Has High Level API	No
highLevelAPI	SDK Repository	

highLevelAPI	Has Domain Specific Compiler	No
Integration		
IP Distribution	Manifest Type	No
IP Distribution	Standalone Simulation	No
IP Distribution	Standalone Simulation SW Requirements	None
IP Distribution	Integration Docs Or Examples	Yes, closed-source proprietary
IP Distribution	IP-XACT	
IP Distribution	System RDL	
Synthesis	Is Synthesizable	Yes
Synthesis	FPGA Synthesis Examples	Yes, closed-source proprietary
Synthesis	ASIC Synthesis Examples	Yes, closed-source proprietary
Simulation	Closed Source Simulation	Yes (Xcelium)
Simulation	Open Source Simulation	No
Evaluation	PPA Results Available	Yes
Physical Implementation		
ASIC Area	Num Gates	4377 GE added to CV32E40S
ASIC Area	On Chip Internal Memory Size	
ASIC Area	Target Clock Frequency	500 MHz (16nm)